Keno: generative noise subtraction for template-free gravitational-wave burst search

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Abstract

Keno is a research software prototype for searching short gravitational-wave bursts without assuming a known waveform shape. A neural network learns to predict detector noise from whitened sensor data; that prediction is subtracted; and the leftover residual is searched with an energy-based statistic, a two-detector timing check, and a simple consistency veto. The prototype is delivered as a full-stack system: a PyTorch/FastAPI inference service, a Node.js orchestration API with caching, and an Angular command center for interactive residual visualization.

On synthetic bursts injected into real LIGO noise, at a 1% false-alarm rate (about one false trigger per hundred noise-only tests), Keno residual search reaches near 100% detection efficiency from signal-to-noise ratio (SNR) 2 to 12. Searching the same data without subtraction, or with a deliberately wrong matched-filter template, is less efficient. On published event times from public catalogs, a coherent two-detector residual search recovers 9 of 18 dual-detector events after rejecting a known glitch near GW170817.

Keno is meant to complement morphology-specific binary black hole classifiers, not replace them. The contribution is a reproducible software path for unmodeled residual search — model, API, and interactive tooling included.

1 Introduction

LIGO and Virgo record a time series called *strain*: a dimensionless measure of how much spacetime stretches as gravitational waves pass the detector. In software terms, strain is a 1D signal sampled at high rate (here 4096 Hz). Before analysis it is usually *whitened*: filtered so that stationary background noise has roughly unit variance across frequency, which makes short anomalies easier to see.

Search methods fall into two complementary families:

- **Morphology-specific / template-based.** If the expected signal shape is known (for example a binary black hole merger chirp), one can match-filter against templates or train a classifier on that family. AresGW [Nousi et al., 2023, Koloniari et al., 2025] is a successful example of the classifier approach for black hole mergers.
- **Template-free / unmodeled.** If the shape is unknown (short rings, noise-like bursts, or other transients), one instead looks for localized excess energy that is consistent across detectors, without assuming a particular waveform.

Keno belongs to the second family. It adds a generative denoising step before the energy search: a neural network predicts instrumental noise $\hat{N}$ and forms the residual

$$R = S_{\text{raw}} - \hat{N}. \tag{1}$$

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Throughout this paper, R denotes that residual (sometimes called "cleaned strain" informally). The residual is then searched with excess power (an STFT peak-energy statistic), an H1+L1 lag scan (Hanford and Livingston detectors), and an envelope-consistency veto that rejects loud single-detector artifacts (*glitches*).

Keno was developed as an independent research prototype by a software engineer. It is not a LIGO collaboration product and is not presented as a production search pipeline. Beyond the model, the project includes a runnable stack for ingestion, inference, evaluation, and interactive inspection.

Relation to morphology-specific classifiers. Systems such as AresGW [Nousi et al., 2023] answer: "does this segment look like a black hole merger I was trained on?" Keno answers: "after I remove the noise I think I understand, does excess energy remain?" Those are different jobs. Keno is intended as a complementary contribution for unmodeled bursts.

Claim. In this prototype, generative subtraction followed by residual excess-power search improves recovery of unknown-morphology bursts at controlled false-alarm rate, relative to (i) excess power on the unsubtracted strain and (ii) matched filtering with a deliberately wrong template.

2 Background terms (for software readers)

SNR Signal-to-noise ratio. Roughly how loud the injected burst is relative to the background; higher SNR is easier to detect.

FAR False-alarm rate. The fraction of pure-noise trials that still cross the detection threshold. A 1% FAR means about 1 in 100 noise-only windows is counted as a false trigger.

Matched filter Correlate the data against a known waveform template. Very powerful when the template matches the true signal; weak when it does not.

Excess power A template-free energy statistic: look for a loud time–frequency patch relative to a robust noise floor (here, the median STFT power).

Residual R What remains after subtracting predicted noise $\hat{N}$ from the input S_{raw}. Ideally background shrinks and a real burst survives.

Glitch A non-astrophysical detector artifact that can look like a burst in one interferometer. Gravity Spy labels common morphological classes (Blip, Koi Fish, Scattered Light, and others).

H1 / L1 LIGO Hanford and LIGO Livingston. Requiring both reduces single-detector false alarms.

GWOSC Gravitational Wave Open Science Center — the public HTTP archive from which Keno downloads open strain frames.

3 Method

Figure 1 summarizes the detection path. Strain segments are whitened and sampled at $f_s = 4096\,\text{Hz}$. Unless noted, analysis windows are $T = 4\,\text{s}$ ($N = 16\,384$ samples).

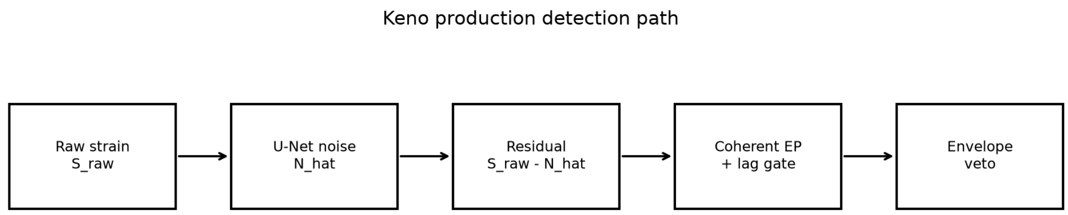

Figure 1: Keno detection path: predict noise $\hat{N}$, form residual $R = S_{\mathrm{raw}} - \hat{N}$, search residual excess power, then check H1+L1 timing consistency with an envelope veto.

3.1 Software stack and interactive command center

Keno is intentionally a full-stack research system, not only a local training script. Beyond the residual statistic itself, the software contribution is a runnable path from open strain to inspected candidate: ingestion, inference, orchestration, and visual audit share one definition of $R = S_{\mathrm{raw}} - \hat{N}$. The architecture has three services:

- **ML engine (Python / FastAPI / PyTorch).** Loads the U-Net checkpoint, fetches and whitens strain via `gwpy`, forms R, and exposes REST endpoints for denoising (`POST /api/v1/denoise`), single-detector detection (`POST /api/v1/detect`), and dual-detector coincidence (`POST /api/v1/detect/coincidence`). Offline evaluation modules (`app.prove`, injection campaigns, glitch stress, catalog follow-up) share the same residual-search code path as the live API.
- **Backend orchestrator (Node.js / Express).** Proxies the Angular UI to the ML engine, forwards GWOSC/offline error details, and caches denoise/detect/coincidence responses in memory. Cold GWOSC downloads can take on the order of 1–3 minutes, so the ML client uses a long inference timeout (300 s); health checks use a short timeout (5 s). Repeat requests for the same GPS window typically return from cache in seconds.
- **Frontend command center (Angular).** An interactive three-pane diff viewer for preset catalog events (e.g. GW150914) or custom GPS times. It plots synchronized raw strain, predicted noise $\hat{N}$, and residual R, and surfaces a “blind search (template-free)” panel: raw vs residual excess power, calibrated thresholds, percent of threshold, and H1+L1 coincidence flags. Synthetic validation mode injects a known burst so a developer can verify the subtraction path without a live download.

Data ingestion from GWOSC is wrapped once in the ML service: open frames are fetched around a GPS timestamp, validated for NaN gaps (offline detectors), whitened, and returned as arrays. Cached background segments used for training and FAR calibration live on disk so evaluation campaigns do not re-download the same quiet windows. The point of this stack is reproducibility and inspection: a physicist or engineer can run the same residual path from a browser or from a CLI prove script.

3.2 Generative noise model

Keno uses a 1D U-Net [Ronneberger et al., 2015] with four encoder/decoder stages (base width 16 channels, kernel size 9, GELU activations, batch normalization). In software terms this is a convolutional autoencoder-style network for 1D sequences: it maps whitened strain S_{raw} to a predicted noise waveform $\hat{N}$. The residual is again

$$R = S_{\mathrm{raw}} - \hat{N}. \tag{2}$$

Training uses cached real public LIGO background windows (GWOSC) for H1/L1. A residual loss term encourages injected short bursts to survive subtraction instead of being absorbed into $\hat{N}$. A hold-out acceptance check requires mean normalized recovery error $\mathrm{RMS}(R - s)/\mathrm{RMS}(s) < 0.15$ on injections with known ground-truth signal s. The checkpoint used here is frozen as `2026-07-paper-v1` (SHA256 prefix `55ce7637...`).

3.3 Template-free excess-power statistic

For a residual or raw series $x(t)$, Keno computes a short-time Fourier transform (Hann window, $n_{\mathrm{perseg}} = 256$, 75% overlap) — the usual STFT used in audio/DSP work. It forms bin power $P(f,t)$, estimates a robust noise floor as median(P), and defines

$$\rho_{\mathrm{EP}}(x) = \max_{f,t} \frac{P(f,t)}{\mathrm{median}(P)}. \tag{3}$$

Intuitively: "how many times louder is the loudest spectrogram cell than the typical background cell?" No template correlation is required.

3.4 Single-detector residual gate

A residual threshold θ_{IFO} is calibrated on noise-only windows at a target FAR. Because imperfect subtraction can create rare large artifacts, the upper 1% of noise-only residual scores is trimmed before taking the $(1 - \mathrm{FAR})$ percentile. In the freeze, $\theta_{\mathrm{IFO}} \approx 4004.6$ at 1% FAR. This gate is useful for single-detector diagnostics. Dual-detector production detection uses the coherent threshold below.

3.5 Coherent dual-detector lag scan

For residuals from Hanford and Livingston, R_{H} and R_{L}, Keno scans small time shifts $|\ell|/f_s \leq 10\,\mathrm{ms}$ (about the light-travel time between sites) and relative sign flips $p \in \{+1, -1\}$:

$$C_{\ell,p} = \frac{R_{\mathrm{H}} + p\,\mathcal{S}_\ell(R_{\mathrm{L}})}{\sqrt{2}}, \tag{4}$$

where $\mathcal{S}_\ell$ is a zero-padded shift. The coherent statistic is

$$\rho_{\mathrm{coh}} = \max_{\ell,p} \rho_{\mathrm{EP}}(C_{\ell,p}), \tag{5}$$

with best lag $\tau^\star = \ell^\star/f_s$. This is a software cross-check: a real astrophysical signal should line up in both detectors within a few milliseconds; a random glitch usually will not.

3.6 Envelope-consistency veto

Independent residual envelope peak times t_{H} and t_{L} are estimated from a short smoothed energy envelope (8 ms boxcar), ignoring the first/last 64 ms so edge artifacts do not dominate. Define

$$\Delta t_{\mathrm{env}} = (t_{\mathrm{L}} - t_{\mathrm{H}}) \times 10^3 \quad (\mathrm{ms}). \tag{6}$$

If $|\Delta t_{\mathrm{env}}| > 50\,\mathrm{ms}$ while the coherent score is loud, the candidate is treated as likely single-detector glitch contamination (for example the known Livingston glitch near GW170817).

3.7 Production detection rule

A dual-detector production trigger is declared if and only if

$$\rho_{\text{coh}} \geq \theta_{\text{coh}}, \tag{7}$$
$$|\tau^\star| \leq 10\,\text{ms}, \tag{8}$$
$$|\Delta t_{\text{env}}| \leq 50\,\text{ms}. \tag{9}$$

The coherent threshold θ_{coh} is calibrated on dual-detector noise trials that already pass the envelope gate. With 1500 noise trials (23 envelope-gated), the 99th percentile of gated coherent EP yields $\theta_{\text{coh}} \approx 173.1$ at empirical joint FAR 0.07% (1/1500). After gating, coherent noise is much quieter than the single-detector residual threshold.

3.8 Baselines

All baselines use the same whitened windows and the same target FAR on noise-only trials:

- **Oracle matched filter** — correlate against the true injected waveform (injection studies only; an upper bound).
- **Mismatched matched filter** — correlate against a fixed sine-Gaussian template that does not match the injected family (control for wrong-template search).
- **Raw excess power** — $\rho_{\text{EP}}(S_{\text{raw}})$ with no generative subtraction.
- **BBH-trained ResNet (control)** — a compact 1D ResNet trained only on black-hole-merger-like chirps versus noise in the same cache, scored as $P(\text{signal})$ on the central 1 s. This is a local morphology-specific control in the spirit of BBH classifiers such as AresGW [Nousi et al., 2023]; it is not a published external weight file.
- **Keno residual EP** — $\rho_{\text{EP}}(R)$.
- **Keno overlap** — cosine similarity between residual and injected signal (evaluation only; needs ground truth).

4 Experiments

Injection campaigns mix short unknown morphologies (sine-Gaussian, ringdown, white-noise burst) into cached real LIGO background at equal FAR. Additional studies include a glitch stress test on public O3 Gravity Spy labels [Gravity Spy et al., 2021], dual-detector noise coincidence FAR, and a follow-up on published GWTC/cWB GPS times (public catalog timestamps). The catalog follow-up is a consistency check, not a discovery claim.

5 Results

5.1 Unknown-morphology efficiency

Figure 2 shows Keno residual EP at 100% efficiency across SNR 2–12 at 1% FAR. Raw excess power on unsubtracted strain is less efficient on the same injections. Mismatched matched filtering remains near ∼20% and never reaches 50% efficiency. The BBH-trained ResNet control, evaluated on these non-BBH bursts, plateaus below saturation — expected for a morphology-specific model outside its training objective.

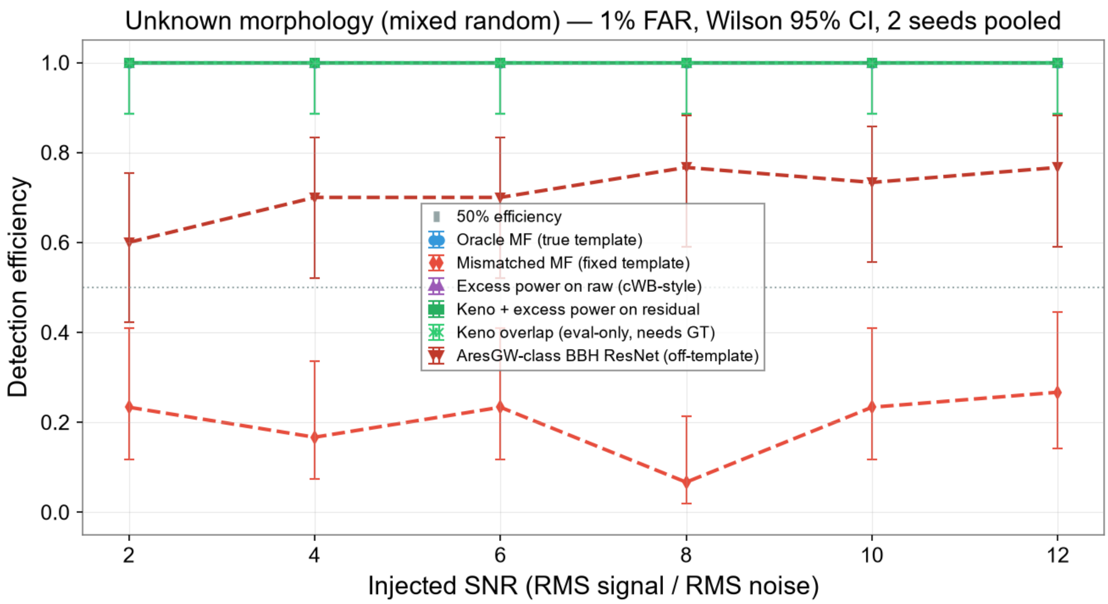

Figure 2: Detection efficiency vs injected SNR at 1% FAR on unknown morphology. Baselines include raw excess power, mismatched matched filtering, and a BBH-trained ResNet control evaluated off-template.

Table 1: Residual survival by Gravity Spy class on the curated O3a stress subset (freeze `2026-07-paper-v1`). Survival means residual EP still exceeds the single-detector threshold after forming $R = S_{\rm raw} - \hat{N}$.

Gravity Spy class	Residual survival	n
Scattered Light	20%	5
Blip	60%	5
Tomte	80%	5
Koi Fish	100%	5
Extremely Loud	100%	5
Whistle	100%	5
All (this freeze)	76.7% (23/30)	30

5.2 Coherent calibration

Envelope gating removes most dual-detector noise outliers. With 1500 noise trials (23 envelope-gated), the coherent EP threshold is 173.1 at empirical joint FAR 0.07%, separate from the single-detector residual threshold ≈ 4005.

5.3 Gravity Spy glitch taxonomy

Single-detector subtraction is *not* a reliable glitch killer. On a curated O3a Gravity Spy subset (30 analyzed triggers across six labels), residual excess power still crossed threshold for 76.7% of glitches (23/30), while the single-detector "kill" rate (raw trigger that disappears after subtraction) was 0/30. Injected unknown-morphology burst controls on the same background were preserved at 40/40 (residual trigger and overlap ≥ 0.75). Table 1 reports residual survival by class (fraction that still trigger after subtraction; $n = 5$ per label in this freeze).

Scattered Light shows the lowest survival: longer, more noise-like morphology that the U-

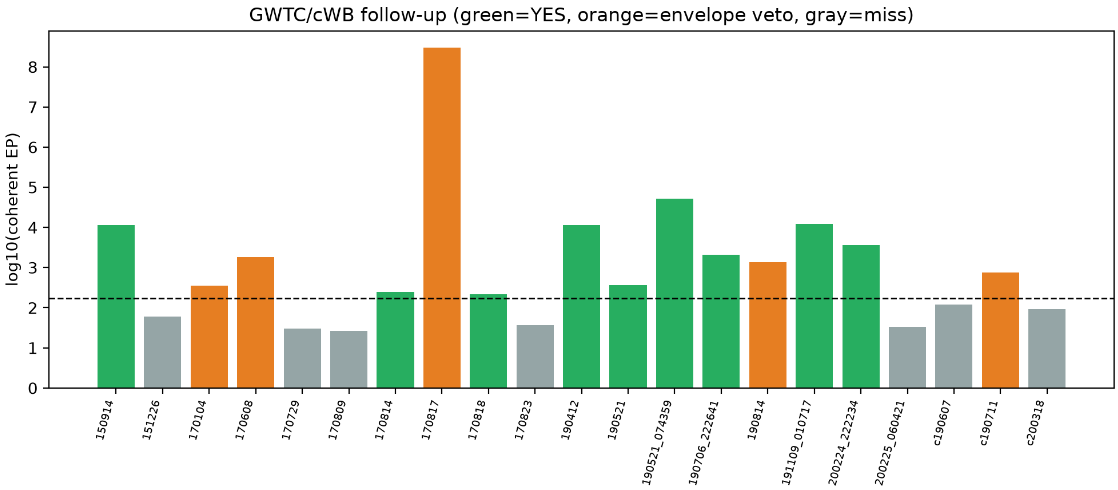

Figure 3: Follow-up coherent EP by event. Green: production YES; orange: envelope veto; gray: miss.

Net tends to absorb into $\hat{N}$. Blips are only partially suppressed. Tomte and the burst-like classes (Koi Fish, Extremely Loud, Whistle) survive in R much as injected astrophysical bursts do. The practical lesson for the software design is explicit: production defense against glitches is multi-detector coincidence and the envelope-timing veto, not an expectation that the U-Net will erase Gravity Spy morphologies on a single interferometer.

5.4 Catalog GPS follow-up

Production coincidence recovers 9/18 dual-detector catalog events (Figure 3). GW150914 is recovered with *Independent*=no and *Production*=yes: Hanford residual EP clears the single-detector gate while Livingston does not, yet the coherent lag scan still passes the dual-IFO coherent threshold with aligned timing (EP $\approx 1.14 \times 10^4$, lag $+8.5$ ms, envelope peak dt -27.6 ms). That asymmetry is expected and is not a failure of the production rule. GW170817 is rejected by the envelope veto despite huge coherent EP (Livingston glitch). Contaminated recoveries are not counted as detections. A small set of cWB-only candidates remains 0/3.

6 Discussion and limitations

Keno is proposed as a complementary software path for unmodeled residual search. Morphology-specific BBH classifiers and matched-filter compact-binary pipelines [Nousi et al., 2023, Koloniari et al., 2025] remain the right tools when the target family is known.

Known limitations of this prototype include:

- weak single-detector glitch rejection, especially for burst-like Gravity Spy classes (Koi Fish, Whistle, Extremely Loud); coincidence and the envelope veto do most of that work;
- residual over-subtraction risk if training is insufficient, mitigated by the recovery-error acceptance check;
- incomplete recovery of some catalog candidates in this freeze;
- cold GWOSC downloads dominate end-to-end latency; cache hits are fast, but first-time analysis of a new GPS window is network-bound;

- development by a single engineer with a research codebase, without collaboration-scale detector characterization or production operations.

7 Reproducibility

Frozen artifacts are stored under `docs/freeze/current/` with checkpoint SHA256 prefix `55ce7637...` (label `2026-07-paper-v1`). Reproduce with:

```
cd ml-engine
python -m app.evaluation.aresgw_class --train
python -m app.prove --freeze --freeze-label 2026-07-paper-v1
```

Interactive inspection uses the Angular command center against the Express backend and FastAPI ML engine (see repository READMEs).

8 Conclusion

Generative noise subtraction followed by template-free residual search can recover unknown-morphology bursts at controlled false-alarm rate in a reproducible research prototype. Keno packages that path as runnable software — inference API, orchestration layer, and visual residual diffs — intended as a complementary contribution for unmodeled residual search alongside existing BBH classifiers, for audit and extension rather than as a finished gravitational-wave pipeline.

Data availability

Code and freeze bundle accompany this manuscript. Large evaluation CSVs are hashed in the freeze manifest.

References

P. Nousi et al., Deep residual networks for gravitational wave detection, Phys. Rev. D **108**, 024022 (2023).

A. E. Koloniari et al., New gravitational wave discoveries enabled by machine learning, Mach. Learn.: Sci. Technol. (2025).

O. Ronneberger, P. Fischer, and T. Brox, U-Net: Convolutional networks for biomedical image segmentation, in Medical Image Computing and Computer-Assisted Intervention (MICCAI), pp. 234–241, Springer (2015).

Gravity Spy Collaboration, Gravity Spy Machine Learning Classifications of LIGO Glitches from Observing Runs O1, O2, O3a, and O3b, Zenodo, doi:10.5281/zenodo.5649212 (2021).